

Summary of Disaster Induced Closures - Raval, Rosenbaum, Wilson

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1: Broad Question

How does natural disasters effect on consumer choice sets impact consumer behavior in switching to new hospitals for care?

2: Specific Question

Given natural disasters are an unexpected cause of hospital closures, how do these closures impact patient hospital choice and what are the implications for consumer demand more broadly? Further, they help to illuminate upon exogenous variation in markets to examine the performance of economic models of consumer behavior.

3: Policy Cares

A policymaker would care because understanding how consumers respond to hospital closures can inform decisions about resource allocation and emergency preparedness in the healthcare sector. If consumers are shown to switch hospitals frequently after a closure, it may indicate a need for better communication and support systems to help patients navigate their options during such events. Additionally, understanding the impact on consumer demand can help policymakers anticipate changes in healthcare utilization patterns and plan accordingly to ensure that healthcare services remain accessible and effective during and after natural disasters.

4: Why Economists Care

Economists would care about this question because it touches on several important economic concepts, including consumer behavior, market dynamics, and the impact of external shocks on supply and demand. Natural disasters represent an exogenous shock that can disrupt established markets, and studying how consumers respond to such disruptions can provide insights into their preferences

and decision-making processes. Additionally, understanding the elasticity of demand for healthcare services in the face of hospital closures can inform broader economic theories related to healthcare markets and the role of government intervention in ensuring access to essential services.

What do we learn from the paper?

This paper provides empirical evidence of the strengths and weaknesses of economic models of consumer behavior in the context of exogenous quasi-experiments. Particularly, we learn that standard models under-report large diversion ratios. Further, they assert that allowance of spatial differentiation and differing specific constants allow for better prediction of diversion ratios. Finally, they assert that greater heterogeneity in consumer characteristics does not lead to greater predicted diversion ratios.

This is all to say that the paper provides a useful framework for understanding how consumers respond to unexpected changes in their choice sets, and highlights the importance of considering spatial differentiation and specific constants in economic models of consumer behavior.

Empirical Strategy

Using likelihood models to recover diversion ratios.

Pitfalls and Overcoming

There may be an endogenous relationship between natural disaster locations and these hospital closures. However, due to the quasi-experimental nature, and their lack of interest in causality, this is not a concern for the authors. There is also a potential issue in assumptions of patient behavior under the circumstances of a natural disaster closing their hospital.

They overcome by using a quasi-experimental design, which allows them to isolate the impact of hospital closures on consumer behavior without being confounded by other factors that may be correlated with both hospital closures and consumer behavior. Additionally, they use a variety of robustness checks and sensitivity analyses to ensure that their results are not driven by any particular assumptions or specifications.